

# Reviewer Pack — Minimal Rank of Hodge Theorems for Polynomial Systems vs DPLL...

Entry 02aaf9954a5e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-24 22:40:58 UTC

## Minimal Rank of Hodge Theorems for Polynomial Systems vs DPLL Refutation Depth

**Entry ID:** 02aaf9954a5e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 22:40:58 UTC

### 1. Conjecture

**Field A** (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Complexity Theory: DPLL Proof Complexity

#### Statement:

[‘For each polynomial system  $P$  with  $n$  variables and  $m$  equations, the minimal rank of the associated Hodge diamond is strictly greater than twice the depth of a DPLL refutation tree for  $P$ .’, ‘Equivalently, for any polynomial system  $P$  with  $n$  variables and  $m$  equations, there exists an embedding into projective space such that the Hodge diamond of its homogeneous coordinate ring has rank at least  $2m$ , where  $m = |P|$ .’, ‘This invariant is preserved under linear substitutions in the defining polynomials.’]

#### Rationale (proposer’s reasoning):

[‘Hodge theory provides a rich algebraic-geometric framework to study the geometry of solutions to polynomial equations. If this geometric structure can be used to bound the complexity of refutation, it may suggest new approaches to proving lower bounds on proof length.’, ‘The Hodge diamond is a combinatorial invariant that captures the topology of the solution space of a polynomial system. It has been studied in algebraic geometry and could potentially provide insights into the computational complexity of related problems.’, ‘DPLL refutation trees are a fundamental object in complexity theory, used to study the proof complexity of propositional logic. A connection between geometric invariants and DPLL tree depth would be a novel bridge between these fields.’]

**Taxonomy category:** GCT\_DET\_PERM (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** 2ae893ad1081d125

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

**Acceptance criterion:**

For each polynomial system  $P$  with  $n$  variables and  $m$  equations, if the minimal rank of the Hodge diamond is greater than twice the depth of the DPLL refutation tree by a margin of at least 3, then support the conjecture; otherwise, falsify it.

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict   | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|-----------|------------|------------------|------------------|
| RELATIVIZATION | UNCERTAIN | 1.00       | HITS             | UNCERTAIN        |
| NATURAL_PROOFS | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |
| ALGEBRIZATION  | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |
| KARP_LIPTON    | SAFE      | 0.50       | UNCERTAIN        | UNCERTAIN        |

### 4. Novelty audit

**Verdict:** NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - (Hodge Theory AND polynomial systems) AND DPLL refutation - (projective space AND Hodge diamond rank) AND (DPLL depth OR proof complexity) - (linear substitutions IN defining polynomials AND Hodge diamond preservation)

**Top relevant hits considered:** - [http://arxiv.org/abs/1204.5159v1] Two simulations about DPLL(T) - [http://arxiv.org/abs/2306.07956v2] Adaptive Monte Carlo Search for Conjecture Refutation in Graph Theory - [http://arxiv.org/abs/1907.07631v1] On the Tour Towards DPLL(MAPF) and Beyond - [http://arxiv.org/abs/1801.10489v3] Hodge level for weighted complete intersections - [http://arxiv.org/abs/0811.2507v2] Cohomology of Substitution Tiling Spaces - [http://arxiv.org/abs/2001.11075v1] A noncombinatorial proof that toric rank 2 bundles on projective space split - [http://arxiv.org/abs/0809.0134v2] Hodge polynomials of some moduli spaces of Coherent Systems - [http://arxiv.org/abs/1006.3462v2] Hodge-Deligne equivariant polynomials and monodromy of hyperplane arrangements

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only,  $\leq 90$ s wall time per cycle **Execution:** rc=1, elapsed=0.1s

#### 5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 10 # Number of variables
    m = 5  # Number of equations

    # Generate a random polynomial system
    coefficients = [[random.randint(-10, 10) for _ in range(n)] for _ in range(m)]
    constant_terms = [random.randint(-10, 10) for _ in range(m)]

    # Compute the Hodge rank (simplified for testing)
    hodge_rank = m
```

```

# Construct a DPLL refutation tree depth
dpll_depth = n * m

# Measure the metric
metric_value = hodge_rank - 2 * dpll_depth

# Check if the conjecture holds
conjecture_holds = metric_value > 3

return {
    "metric_name": "Hodge Rank vs DPLL Depth",
    "metric_value": metric_value,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"n={n}, m={m}, hodge_rank={hodge_rank}, dpll_depth={dpll_depth}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        result = run_trial(seed)
        results.append(result)

        print(f"TRIAL: {result}")

    # Compute mean/std of metric_value and fraction of seeds where conjecture_holds
    if all(r["conjecture_holds"] for r in results):
        support_fraction = 1.0
    else:
        support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={sum(r['metric_value'] for r in results)/len(results)} std={sum((r['metric_value'] - sum(r['metric_value'] for r in results)/len(results))**2 for r in results)/len(results)}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

```

0'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}
TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'n=2, m=3, hodge_rank=-95, dpll_depth=6'}

```

```

TRIAL: {'metric_name': 'Hodge Rank vs DPLL Depth', 'metric_value': -95, 'instances_tested': 1, 'conje
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--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_albc98f9.py", line 69, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_albc98f9.py", line 69, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^

```

## 8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

Safety rail: critic\_challenge\_falsified | original: The counterexample provided shows that for a polynomial system with  $n=10$  and  $m=5$ , the Hodge rank (5) is not strictly greater than twice the DPLL refut | next: Investigate polynomial systems where the Hodge rank is close to twice the DPLL refutation depth but still violates the conjecture's condition. Additionally, explore alternative methods for proving or disproving the conjecture.

## 11. Audit log (LLM calls)

**Total LLM calls:** 9

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | ollama_remote | glm4:latest      | 0         | 0   | 12899        |
| 2 | preregistration | ollama_remote | glm4:latest      | 0         | 0   | 6083         |
| 3 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 4640         |
| 4 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 5738         |
| 5 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 12036        |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 8487         |
| 7 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 10149        |
| 8 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 7120         |
| 9 | judge           | ollama_remote | glm4:latest      | 0         | 0   | 12597        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 79748 ms total latency. Provider mix: {'ollama\_remote': 9}

(full prompt+response transcripts available in `research/audit/02aaf9954a5e.jsonl`)

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71

2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/02aaf9954a5e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** `research/pvsnp_sandbox/test_*.py` (per-cycle)

**Replay tarball:** `research/replay/02aaf9954a5e.tar.gz` (if generated)